

CASE-02

A six-point drop in delivery success is not a delivery problem.

Diagnosing a fall from 85% to 79% first-attempt success out of raw order data — then designing the system that recovers it.

SELF-DIRECTED CASE STUDY

Nabil Hakeem

Data diagnosis · System design · Roadmap

linkedin.com/in/nabil-hakeem-pm

The failure isn't where it's labelled.

Across roughly 95,000 orders in nineteen statuses, only 31 records are explicitly marked “delivery failed”. Meanwhile **12,850 orders completed a return**. The loss doesn't live in the column named after it.

Reading the distribution rather than the labels, the causes cluster in four places: address and contact data quality, customer availability at the delivery moment, operational routing and driver-customer contact, and the absence of any address validation at order time.

None of those are solved by attempting the delivery again.

From the order data

ORDERS DELIVERED

57,757

RETURNS COMPLETED

12,850

SUCCESS RATE

85% → 79%

Whose problem is this?

The framing you choose here determines every line of the roadmap that follows.

TREAT IT AS A DRIVER PROBLEM

More attempts, more cost per drop, and the same failure rate — because the second van goes to the same wrong address.

TREAT IT AS AN INFORMATION PROBLEM

Either the address was wrong or the customer was never told when to be home. Both are fixable before the van leaves.

I proposed a **Dynamic Delivery Coordination System** — real-time coordination between customer, driver and system — and sequenced it strictly by where the failure originates.

What I proposed, and why.

1 • Validate the address at checkout, not at the door

Geolocation pin plus third-party validation during order placement. The cheapest failed delivery is the one prevented before dispatch.

2 • Let the customer pick the window

Slots generated dynamically against real driver capacity and geography — not a static list that promises times the operation can't hold.

3 • Make a failed delivery recoverable in one tap

Reschedule directly from the failure notification, so the recovery path starts at the moment of frustration instead of a support queue.

4 • Route optimisation last, not first

It genuinely improves cost per drop. It does nothing about a wrong address — so it earns the third slot, not the headline.

I sequenced address validation first and AI route optimisation last.

Route optimisation is the impressive item on this list, and the one a stakeholder reacts to. But it improves the efficiency of journeys to addresses that may not exist. Validation is dull, cheap, and sits upstream of the actual failure.

Targets, one quarter

Success 79% → 85%

Failed 21% → 15%

Reschedule success 95%

Address errors -30%

What I took from it

The most impressive component is rarely the highest-leverage one. Ordering a roadmap by where the failure originates beats ordering it by how advanced each piece sounds in the readout.

Happy to talk through the thinking —
[linkedin.com/in/nabil-hakeem-pm](https://www.linkedin.com/in/nabil-hakeem-pm)